

How I Completed Exercise 01 On a Mac using Remote Editing

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Remote Editing

Some editors can access files that are stored remotely and let you edit them the same way you would a file on your own computer. Whenever the editor needs to read or write the file, it uses the network to access the file on a remote system. I've done this with BBEdit (MacOS), emacs (Linux, MacOS or Windows) and Sublime Text (Linux, MacOS or Windows). This is easier to set up and use on BBEdit. That's the approach these instructions describe.

If you want to try working this way, you'll need to install BBEdit On a Mac, I was able to install BBEdit from the app store. For Sublime Text, I was able to download it from their homepage at <https://www.sublimetext.com>. The specifics of how to access a remote file will vary by editor, but the overall process is basically the same.

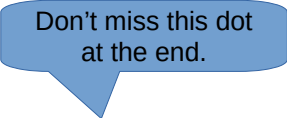
Completing the Exercise

You'll be running the compiler and executing your programs on a remote Linux system at NCSU. You'll just be editing your files using an editor that's running locally. First, login to an EOS remote linux machine. On MacOS, you should be able to do this using ssh from a terminal window (using your actual unity ID below):

```
$ ssh unity-id@remote.eos.ncsu.edu
```

Once you're logged in to a remote Linux machine, set up a directory where you're going to work. I'd make a directory for CSC 230 exercises, then maybe a subdirectory for each particular exercise. Then, you can use **cp** to copy the files you need from our shared CSC230 directory. I was able to just copy-and-paste the **cp** command from up near the top of the exercise.

```
$ mkdir ex230
$ cd ex230
$ mkdir 01
$ cd 01
$ cp /mnt/coe/workspace/csc/CSC230/ex/exercise01/* .
```



Don't miss this dot
at the end.

Then, you can run BBEdit on your local system (e.g., your laptop). From the editor, you can remotely open the files you just copied on the remote Linux system.

- In your login window for the remote Linux machine, type **echo \$HOME**. This should print out your home directory. You'll need it below.
- From the File menu, choose "Open from FTP/SFTP Server ...". This will pop up a little window for connecting.
- Choose **remote.eos.ncsu.edu** as the server and choose SFTP (with the default port, 22). Type in your unity ID and your NCSU password then connect.

- In the Path field, enter the path to your home directory in AFS (the output from the echo command above).
- After a little delay, this should bring up a window for selecting files. Open the arithmetic.c file you downloaded to AFS. Based on the path I used above, you would need to select ex230, then 01, then arithmetic.c.

In my editor, I added the missing lines of code, then saved the file. This should save our edited file in the remote file system on remote.eos.ncsu.edu, rather than saving it to your local machine. You can check by typing the following on the remote Linux machine. This should show the contents of your edited source file right in the terminal:

```
$ cat arithmetic.c
```

Then, try compiling your edited source file using the typical compiler options:

```
$ gcc -Wall -std=c99 arithmetic.c -o arithmetic
```

If my program hadn't compiled cleanly, I'd have gone back to the editor and fixed whatever was wrong. Since it compiled OK, I'm ready to try it out. In shell code below, I'm using the tee command to send output to the terminal window, while still capturing a copy of the output to a file. This is handy for seeing the output and still capturing a copy to a file so we can later check it against the expected output. After running the arithmetic program, I check its exit status to make sure it's zero, then I compare the output against the expected output file.

```
$ ./arithmetic | tee output.txt
500000000500000000
$ echo $?
0
$ diff output.txt expected.txt
```

Both of these tests look good. Since I was running this on an EOS Linux machine, it should work the same on any common platform system. I should be ready to turn it in.

Submitting a Solution

You'll need to use a web browser to submit your solution to the exercise. This will be easier to do from your local system. Since your file is still on the remote Linux machine, you'll need to download a copy first. In your terminal window, you can exit from the remote linux machine, then use **sftp** to download a copy of your source file. Or, if you're accustomed to a graphical file transfer program like FileZilla, you can use that instead. From the terminal, the following commands connect using sftp (use your actual unity ID here), then change directory to wherever you worked on the exercise, then use get to pull down a copy of your modified source code.

```
$ sftp unity-id@remote.eos.ncsu.edu
Connecting to remote.eos.ncsu.edu...
Password: *****
sftp> cd ex230
sftp> cd 01
```

```
sftp> get arithmetic.c  
Fetching ../ex230/01/arithmetic.c to arithmetic.c  
sftp> exit
```

Start up a web browser on your local system and visit the Moodle page for CSC 230. Clicking on the assignment for exercise 01 should take you to a submission page where you can chose "Add Submission". From there, you can drag a copy of your solution into the submission box. Select your arithmetic.c file for submission then choose "Save changes" down near the bottom of the web page.